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Course: ECE 5210

Subject: Lab 4, Decimation and 60 Hz noise

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1 Introduction

The purpose of this lab is to implement an FIR filter that will remove a certain frequency from a signal. In this case, the frequency to be removed is a 60Hz power source signal from a cardiac signal. This lab will show that trying to notch a certain frequency with a 3 tap FIR filter without down sampling is not really possible. In order to make the notching work correctly, signal will be down sampled, processed, up sampled, and then interpolated to remove the unwanted noise.

2 Theory

2.1 FIR Notch Filtering

The basic equation for the 3 tap notching filter is shown below in equation 1.

$$h[n] = \delta[n] + a\delta[n-1] + b\delta[n-2] \tag{1}$$

The filter coefficients are calculated for a notch filter at 60 Hz using a 48 kHz sampling rate and the result is shown below

$$h[n] = \delta[n] - 2cos(0.0025\pi)\delta[n-1] + \delta[n-2]\} \tag{2}$$

2.2 Under-sampled FIR Notch Filtering

In order to get the desired filter, the sampling frequency needs to be lowered. The board samples at 48kHz, so that won't be changed. In order to simulate sampling at a lower rate, the data will be down sampled with M = 100. This means that 1/100 samples will be kept, and the rest will be discarded.

After down-sampling, the data will be run through an FIR Filter that corresponds with this under-sampled approach. The filter coefficients are recomputed with down-sampled approach. The digital frequency when sampled is $\pi/4$. The equation for these new coefficients is shown below

$$h[n] = \delta[n] - 2cos(0.25\pi)\delta[n-1] + \delta[n-2]\} \tag{3}$$

After processing the data with this FIR filter, it will be upsampled and then interpolated. The interpolation used here will be linear interpolation. The equation for the interpolation is shown below

$$y = y_1 + (x)\frac{y_2 - y_1}{100} \tag{4}$$

3 Results

3.1 FIR Notch Filtering

The DTFT of the FIR filter is plotted in figure 1.

When the data from this filter is analyzed, it becomes obvious that this is a pretty lousy filter. The ω that ends up getting cut out is very close to zero, because the data is being sampled too fast

3.2 Under-sampled FIR Filtering

The DTFT of the downsampled FIR filter is plotted in figure 2. This DTFT shows that the downsampled approach works much better.

3.3 Frequency Response

The theoretical response, and the measured response from the board are plotted on the same plot in Figure 3. It shows that the response aligns exactly until about 200 Hz. This is because that is when aliasing kicks in.

3.4 Filter Passthrough/Processing

A cardiac signal that is corrupted by a 60 Hz signal is sent through the board and the passthrough and processed data is plotted in Figure 4. It shows that the filtering is effective in removing the 60 Hz noise

4 Discussion and Conclusions

This lab gave me a good understanding of implementing an FIR filter using downsampling. At first I was confused on how this worked in real time, because I can't add data back into negative time. But then I realized that we just delay the sampled by 100 and start outputing them when we have enough data to start interpolation.

This lab was a little harder than the other labs before. I got stuck on indexing incorrectly for a while on this lab. I decided to break down the lab into smaller, testable parts to be able to debug where I was going wrong. By testing smaller components and seeing what worked, and what didn't work, I was able to complete this lab as needed.

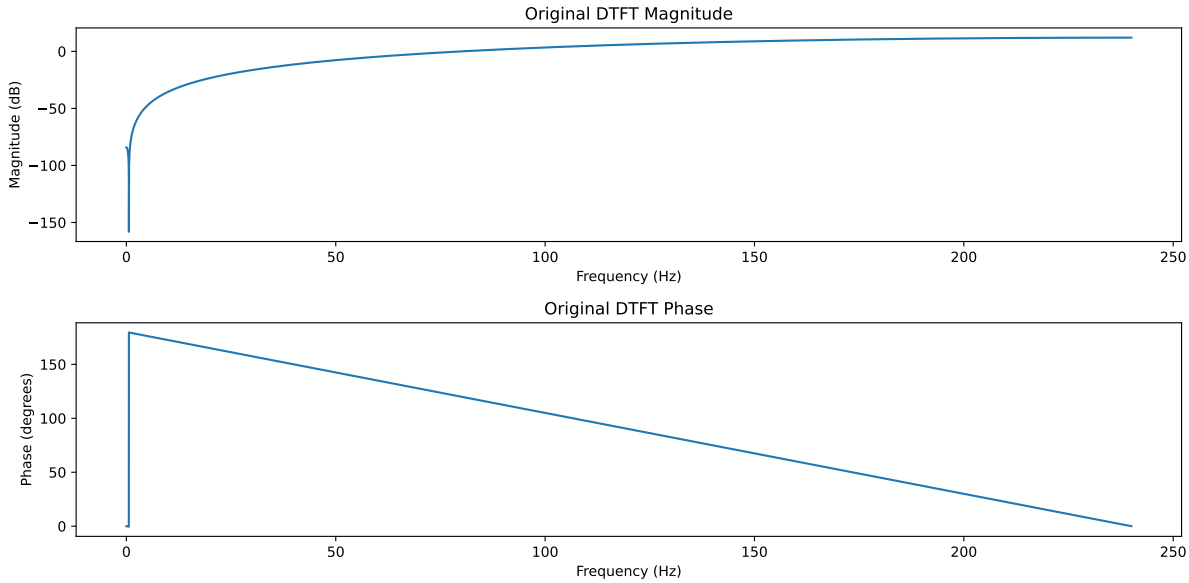

Figure 1: FIR Notch Filter

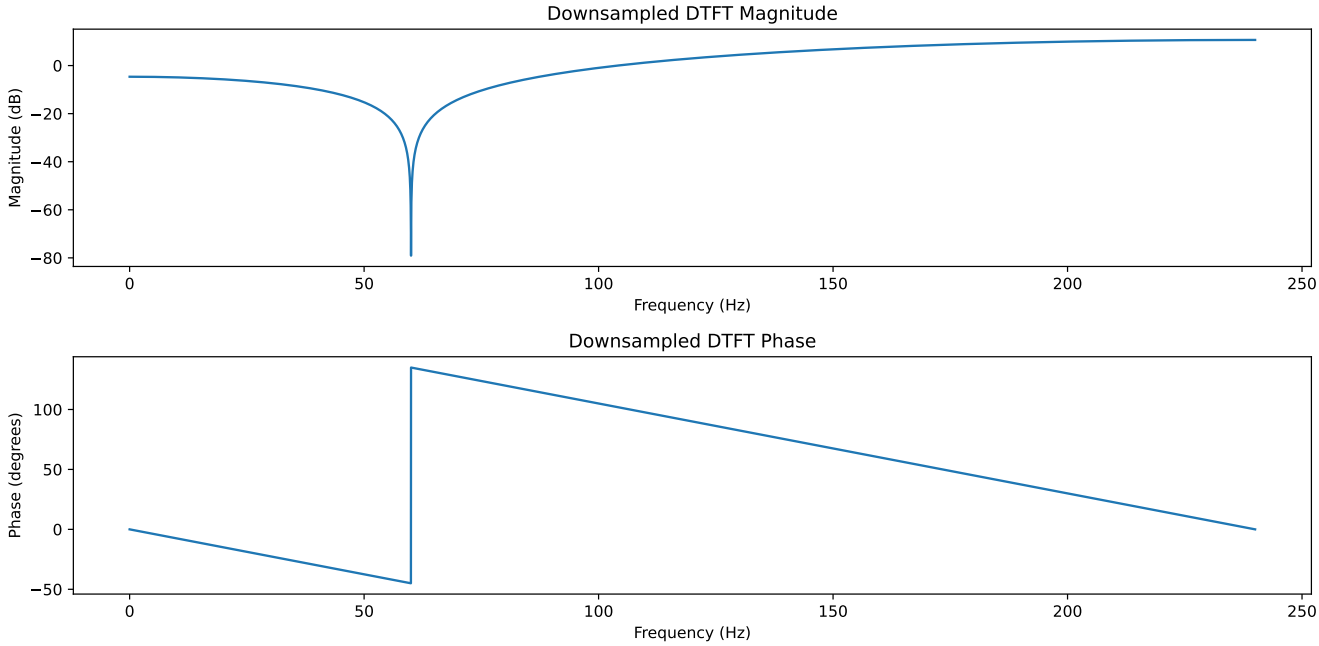

Figure 2: Downsampled FIR Notch Filter

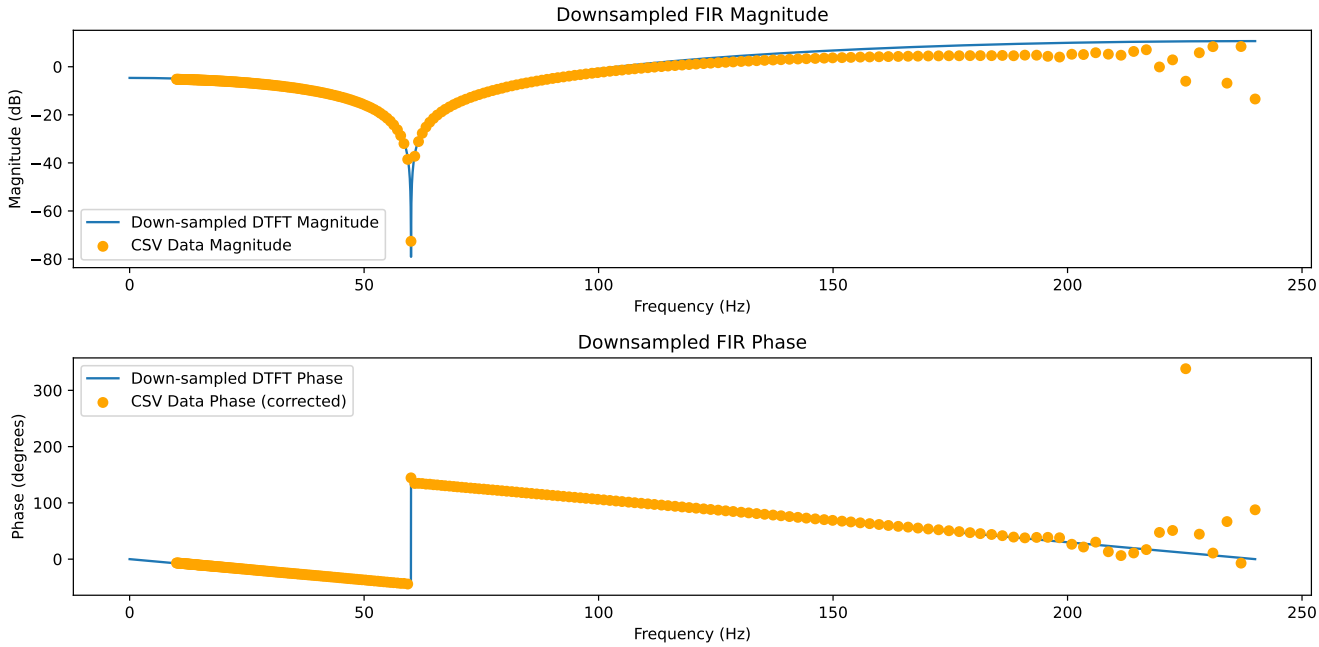

Figure 3: Frequency Response Plot

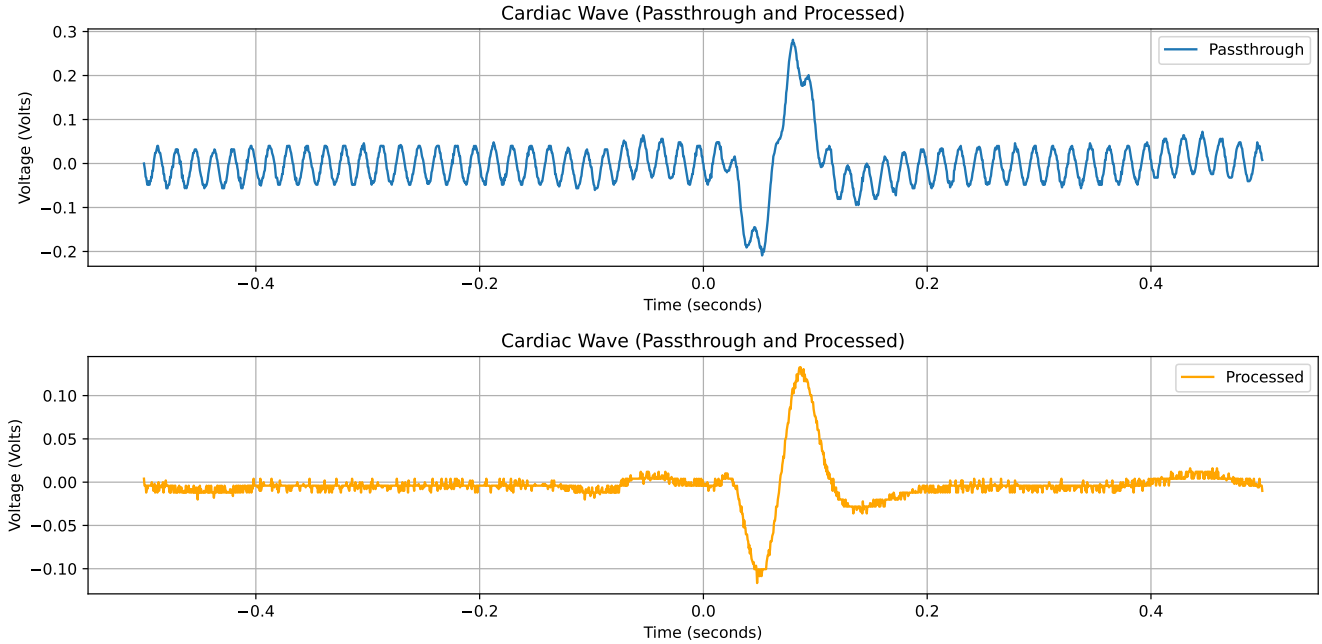

Figure 4: Interference Removal Plot